

Q1. Solution:

$$V = k \int \frac{\rho_L d\ell'}{|\vec{r} - \vec{r}'|}$$

$$\vec{r} = h\hat{a}_z = (0, 0, h)$$

$$\vec{r}' = a\hat{\phi}$$

$$|\vec{r} - \vec{r}'| = \sqrt{h^2 + a^2}$$

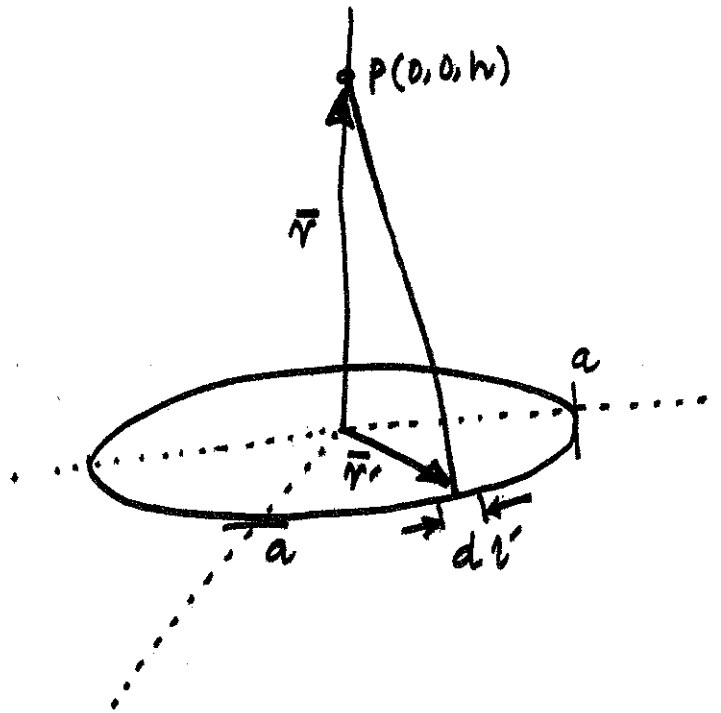
Note that $d\ell' = a d\phi$

$$V(\vec{r}) = k \int \frac{\rho_L a d\phi}{\sqrt{h^2 + a^2}} = \frac{k \rho_L a}{\sqrt{h^2 + a^2}} \int_0^{2\pi} d\phi = \frac{k \rho_L 2\pi a}{\sqrt{h^2 + a^2}}$$

Denoting $Q = 2\pi a \rho_L$ as total amount of charge on the circular ring, we obtain

$$V(\vec{r}) = \frac{k Q}{\sqrt{a^2 + h^2}}.$$

PROVED.



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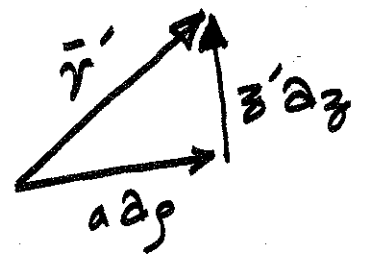
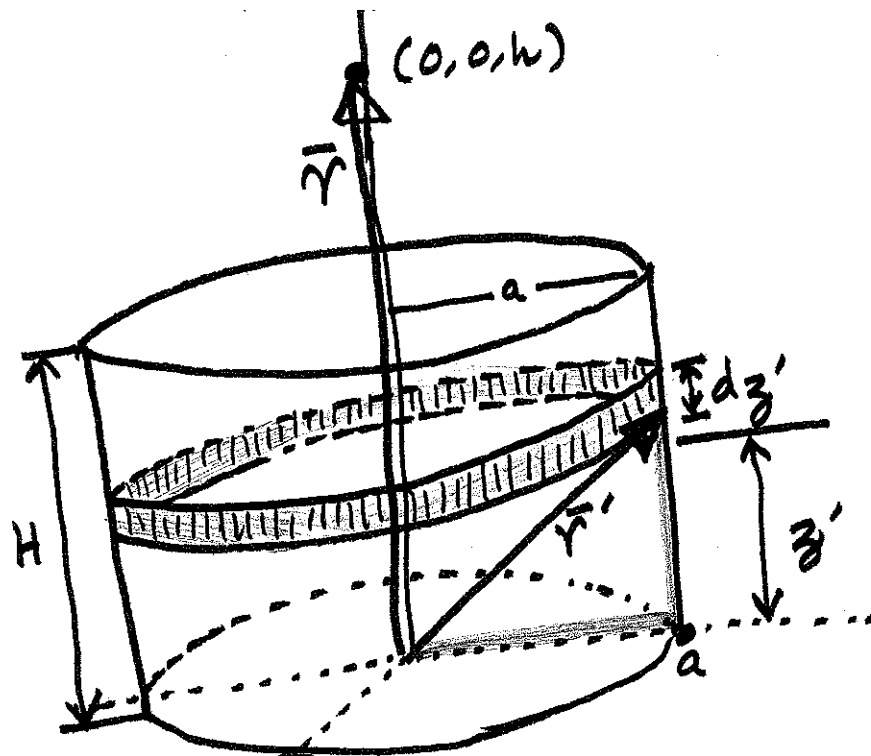
Q2. Solution:

Note that the vector $\vec{r}'$ has two components.

$$\vec{r}' = a \hat{a}_\rho + z' \hat{a}_z$$

while

$$\vec{r} = (0, 0, h) = h \hat{a}_z$$



So

$$|\vec{r} - \vec{r}'| = \sqrt{a^2 + (h - z')^2}$$

$$V(\vec{r}) = V(0, 0, h) = k \int \frac{\rho_s ds'}{|\vec{r} - \vec{r}'|}$$

$ds' = (2\pi a) dz' = \text{area of the shaded region}$

$$V(\vec{r}) = k \rho_s 2\pi a \int_0^h \frac{dz'}{\sqrt{a^2 + (h - z')^2}}$$

$$= k \rho_s 2\pi a \left[-\sinh^{-1} \left(\frac{h - z'}{a} \right) \right]_0^h$$

$$= k \rho_s 2\pi a \left[\sinh^{-1} \left(\frac{h}{a} \right) - \sinh^{-1} \left(\frac{h - h}{a} \right) \right]$$

Q.E.D.

Q3. Solution:

$$\vec{E} = \frac{5z^2}{\epsilon_0} \hat{a}_z \text{ V/m.}$$

$$\Rightarrow \vec{D} = \epsilon_0 \vec{E} = 5z^2 \hat{a}_z \text{ C/m}^2$$

$$\rho_v = \nabla \cdot \vec{D} = \frac{\partial}{\partial z} 5z^2 = 10z \text{ C/m}^3$$

$$\begin{aligned} \text{Total charge} &= \int \rho_v dv \\ &= \int_{-2}^2 \int_{-2}^2 \int_{-2}^2 10z dx dy dz \end{aligned}$$

$$= 4 \times 4 \times 10 \int_{-2}^2 z dz$$

$$= \frac{160}{2} z^2 \Big|_{-2}^2$$

$$= 80(4 - 4) = 0 \text{ C.}$$

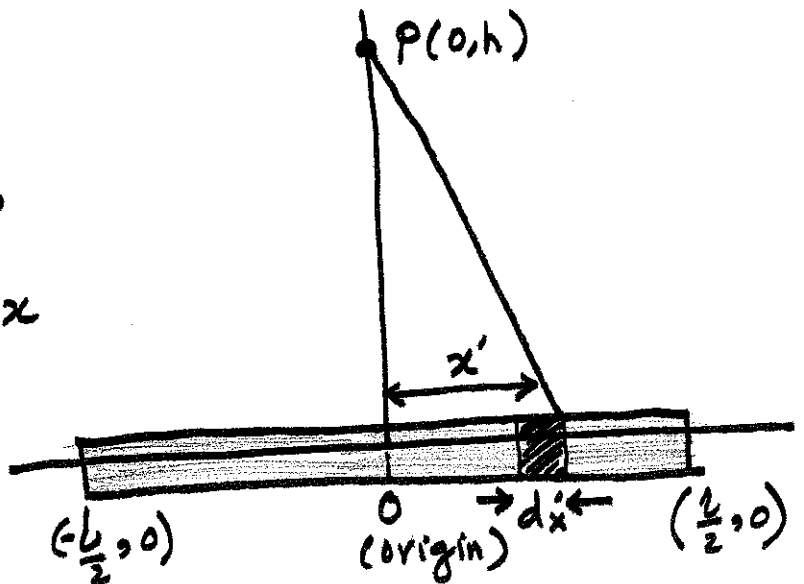
Q.E.D.

Q4. Solution

Note $\vec{r} = (0, h) = h \hat{a}_y$

$\vec{r}' = (x', 0) = x' \hat{a}_x$

$$|\vec{r} - \vec{r}'| = \sqrt{h^2 + (x')^2}$$



$$V(\vec{r}) = V(0, h)$$

$$= k \int \frac{\rho_l dx'}{|\vec{r} - \vec{r}'|}$$

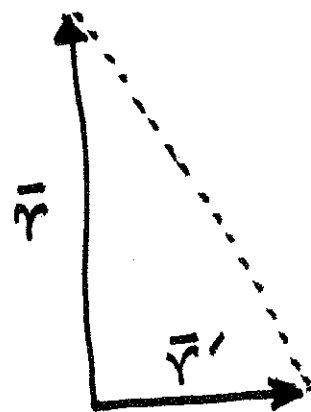
$$= k \rho_l \int_{-l/2}^{l/2} \frac{dx'}{\sqrt{h^2 + (x')^2}}$$

$$= 2k \rho_l \int_0^{l/2} \frac{dx'}{\sqrt{h^2 + (x')^2}}$$

$$= 2k \rho_l \log(x' + \sqrt{x'^2 + h^2}) \Big|_0^{l/2}$$

$$= 2k \rho_l \left[\log\left(\frac{l}{2} + \sqrt{\frac{l^2}{4} + h^2}\right) - \log h \right]$$

Q.E.D.



Q5. Solution.

$$a) Q = \int_0^L \int_0^{2\pi} \int_{0.001}^{\rho_i} 4\rho \cdot \rho d\rho d\phi dz = \frac{8\pi L}{3} [\rho_i^3 - 10^{-9}] \mu C$$

$$b) Q = \oint \vec{D} \cdot d\vec{S} = D_\rho (2\pi \rho_i L)$$

$$D_\rho = \frac{Q_0}{2\pi \rho_i L} = \frac{4(\rho_i^3 - 10^{-9})}{3 \rho_i}.$$

c) Do yourself.